Wealth Inequality and Safe Asset Demand

Ding & Jiang *WP* 2026

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Motivation

Huggett economies:

- Uninsurable idiosyncratic risk creates precautionary savings
- This lowers the risk-free rate
- And shapes wealth inequalities

This paper: Which consequences for asset pricing? Can the SDF be pushed low enough to support asset price bubbles?

Overview of results

- Permanent risk: some hhs. become much poorer than the average$\rightarrow$ asset cash flow is much larger than their wealth $\rightarrow$ valuation bubble
- Stationary settting: if the left tail of the distribution is thick enough, this possibility remains
- This is a quantitative question: requires a careful analysis of the shape of the wealth distribution, the left tail also matters!

Setting

$$\max_{c_t^i} \mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} \log(c_t^i) dt \right] \text{ s.t. idiosyncratic risk}$$

$\implies$ equilibrium representation: $\frac{dc_t^i}{c_t^i} = g_t dt + \sigma_t dZ_t + \tilde{\sigma}_t d\tilde{Z}_t^i$ nests many HA settings

Notice that c_t is not bounded away from zero as $c_t \sim$ Log-$\mathcal{N}$

Yields the familiar results:

$$m_t^i = e^{-\rho t} \frac{1}{\rho n_t^i}, \quad \frac{dm_t^i}{m_t^i} = -r_t^f dt - \sigma_t dZ_t - \tilde{\sigma}_t d\tilde{Z}_t^i$$

Which implies that the risk-free rate is

$$r_t^f = \rho + g_t - \sigma_t^2 - \tilde{\sigma}_t^2$$

Where we see the Huggett (1993) result: $\uparrow$ volatility$\rightarrow\uparrow$ precau. savings $\rightarrow \downarrow$ risk-free rate

Individual and aggregate wealth valuation

Individual wealth dynamics:

$$dn_t^i = -c_t^i dt + n_i^t dr_t^{w,i} \implies \text{IBC: } m_t^i n_t^i = \mathbb{E}_t\left[\int_t^\infty m_s^i c_s^i ds\right] + \lim_{T\to\infty} \mathbb{E}_t[m_T^i n_T^i]$$

Dynamic programming implied TVC:

$$\lim_{T\to\infty} \mathbb{E}_t[m_T^i n_T^i] = 0$$

This TVC always holds as $m_t^i n_t^i \propto e^{-\rho t}$

Aggregate wealth dynamics:

$$\frac{d\bar{n}_t}{\bar{n}_t} = g_t dt + \sigma_t dZ_t$$

Aggregate "TVC"

$$\lim_{T\to\infty} \mathbb{E}_t[m_T^i \bar{n}_T] = 0 \iff \lim_{T\to\infty} \mathbb{E}_t[\bar{m}_T \bar{n}_T] = 0 \ \textbf{?}$$

Two perspectives on aggregate valuations

A distributional perspective:

Let $\eta_t^i = \frac{n_t^i}{\bar{n}_t}$, we have

$$\mathbb{E}_0\left[\bar{m}_T \bar{n}_T\right] = e^{-\rho T}\mathbb{E}_0\left[\int_0^1 \frac{di}{\eta_T^i}\right]$$

If for a large share of households $\eta_T^i \to 0$ sufficiently fast then the aggregate "TVC" may fail

Wealth share dynamics:

$$\frac{d\eta_t^i}{\eta_i^t} = \tilde{\sigma}_t d\tilde{Z}_t^i$$

We have $\eta_t^i \to 0$ *a.s.* even if $\mathbb{E}[\eta_t^i] = \int_0^1 \frac{n_t^i}{\bar{n}_t} di = 1$

$\hookrightarrow$ Almost all households become increasingly poor, a negligible share gets rich and carries the mean

Result: the aggregate "TVC" fails if $\eta_T^i \to 0$ faster than time discounting

Two perspectives on aggregate valuations

The "r-g" perspective:

Discounted aggregate wealth behaves as

$$\bar{m}_t \bar{n}_t \propto e^{-(r_t^f + \sigma_t^2 - g_t)} = e^{-\rho + \tilde{\sigma}_t^2}$$

Same result: the aggregate TVC is violated when the magnitude of idiosyncratic risk is sufficiently large relative to the subjective discount rate ρ

Because inequalities explode, aggregate wealth is discounted slower than it grows

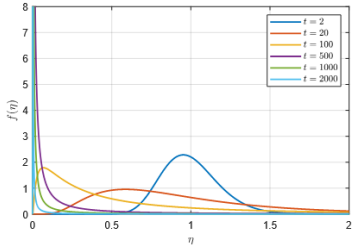
From the point of view of an asset issuer:

Faces an IBC

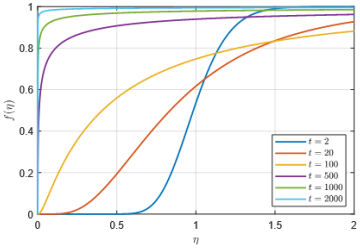
$$m_t^i b_t = \mathbb{E}_t \left[\int_0^\infty m_{t+j}^i s_{t+j} \, dj \right] + \lim_{T \to \infty} \mathbb{E}_t \left[m_T^i b_T \right]$$

$\hookrightarrow$ perpetual deficits can be matched by the bubble term or the asset can have infinite present value

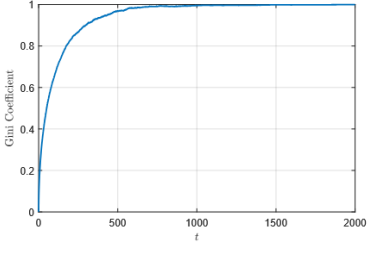
Simulation: $\gamma = 2, \sigma_t = 0, g_t = 0, \tilde{\sigma}_t = \tilde{\sigma}$



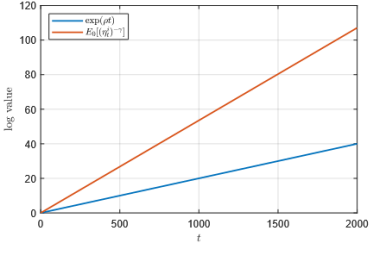
(a) Distribution of Wealth Share.



(b) Cumulative Distribution of Wealth Share.



(c) Gini Coefficient.



(d) TVC for Aggregate Wealth.

Transitory idiosyncratic risk

Can the mechanism survive in a stationary setting?

Mean reversion in wealth: individual wealth is reset to the average level at a Poisson rate ζ

$$\frac{c_t^i}{n_t^i} = \rho + \zeta, \quad \frac{dc_t^i}{c_t^i} = g_t dt + \sigma_t dZ_t + \tilde{\sigma} d\tilde{Z}_t^i + \frac{\bar{n}_t - n_t^i}{n_t^i} d\tilde{J}_t^i$$

$$r_t^f = \rho + \zeta + g_t - \sigma_t^2 - \tilde{\sigma}_t^2$$

$\hookrightarrow$ households become effectively more impatient as jump risk erases accumulated wealth

Get the following average SDF dynamics

$$d\bar{m}_t = -(\rho + g_t - \sigma_t^2 - \tilde{\sigma}^2)\bar{m}_t dt - \sigma_t \bar{m}_t dZ_t + \int_0^1 e^{-\rho t} \frac{d\tilde{J}_t^i}{\rho(\bar{n}_t - n_t^i) m_0^i} di$$

For an asset issuer this implies

$$b_0 = \mathbb{E}_0 \left[\int_0^T \bar{m}_t s_t dt \right] + \mathbb{E}_0 \left[\bar{m}_T b_T \right] - \zeta \mathbb{E}_0 \left[\int_0^T e^{-\rho t} \frac{\bar{n}_0}{\bar{n}_t} b_t dt \right] \leq \mathbb{E}_0 \left[\int_0^T \bar{m}_t s_t dt \right] + \mathbb{E}_0 \left[\bar{m}_T \bar{n}_T \right]$$

The distributional perspective

The stationary distribution of wealth shares features two Pareto tails

$$f(\eta) = C\eta^{-\frac{3}{2}+\frac{\sqrt{1+\frac{8\zeta}{\tilde{\sigma}^2}}}{2}} H(\eta)H(1-\eta) + C\eta^{C}\eta^{-\frac{3}{2}-\frac{\sqrt{1+\frac{8\zeta}{\tilde{\sigma}^2}}}{2}} H(\eta-1)$$

$\hookrightarrow \uparrow \tilde{\sigma}$ heavier tails, $\uparrow \zeta$ lighter tails.

With a general risk aversion coefficient γ:

$$\lim_{T\to\infty} \mathbb{E}_0\left[\bar{m}_T \bar{\eta}_T\right] = \lim_{T\to\infty} e^{-\rho T}\left(\frac{\bar{n}_T}{\bar{n}_0}\right)^{1-\gamma} \mathbb{E}_0\left[\int_0^1 \left(\eta_T^i\right)^{-\gamma} di\right]$$

The last term is finite if $\zeta > \frac{\gamma(1+\gamma)}{2}\tilde{\sigma}^2 \rightarrow$ If jump intensity is strong enough, then the aggregate "TVC" holds

Result: Aggregate TVC fails if $0 < \hat{c} \leq \gamma\tilde{\sigma}^2$ where $\hat{c} = \frac{c_t}{n_t} = \hat{c}(S_t; \rho, \zeta)$.

In BMS(2024) $\hat{c}(\vartheta_t; \rho, \zeta)$ solves $\hat{c} - (\gamma-1)\Phi(\iota) = \rho + \zeta - (\gamma-1)\delta - \frac{\gamma(\gamma-1)}{2}\bar{\chi}^2(1-\vartheta)^2\tilde{\nu}^2$

Summary

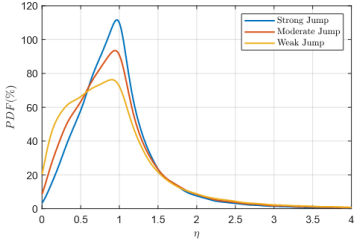
(a) With Wealth Reset, Transitory Idiosyncratic Risk

Jump Intensity	Parameter Region	TVC holds?	$\mathbb{E}[\int(\eta^i_\infty)^{-\gamma}di]=\infty$?	$r>g$?
High	$\hat{c}>\gamma\tilde{\sigma}^2+\rho+(\gamma-1)g$	Yes	No	Yes
Moderate	$\gamma\tilde{\sigma}^2<\hat{c}\leq\bar{\gamma}\bar{\sigma}^2+\rho+(\gamma-1)g$	Yes	Yes	Yes
Low	$0<\hat{c}\leq\gamma\tilde{\sigma}^2$	No	Yes	No

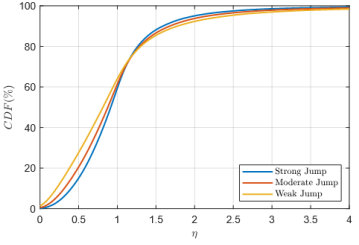
(b) Without Wealth Reset, Permanent Idiosyncratic Risk

Idiosyncratic Volatility	Parameter Region	TVC holds?	$\mathbb{E}[\int(\eta^i_\infty)^{-\gamma}di]=\infty$?	$r>g$?
Low	$\hat{c}>\gamma\tilde{\sigma}^2$	Yes	Yes	Yes
High	$\hat{c}\leq\gamma\tilde{\sigma}^2$	No	Yes	No

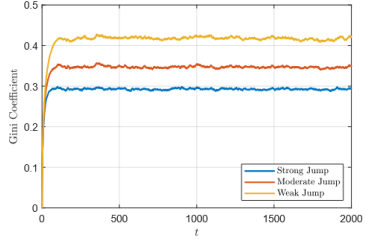
Illustration: $\zeta \in \{5.4\%, 3.8\%, 2.5\%\}$



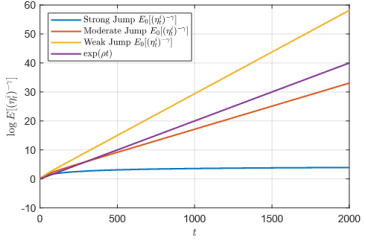
(a) Distribution of Wealth Share.



(b) Cumulative Distribution of Wealth Share.



(c) Gini Coefficient.



(d) TVC for Aggregate Wealth.

The "$r - g$" perspective

Back to log utility, discounted wealth has dynamics

$$d[\bar{m}_t \bar{\eta}_t] = -\left[\rho + \zeta - \tilde{\sigma}_t^2\right] \bar{m}_t \bar{\eta}_t dt + \zeta e^{-\rho t} \bar{\eta}_0 dt$$

Thus the aggregate "TVC" holds if

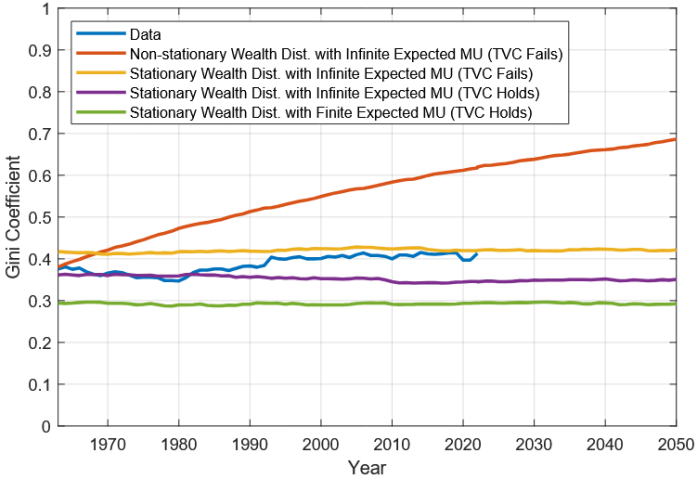
$$\rho + \zeta > \tilde{\sigma}^2 \Longleftrightarrow r_t^f + \sigma^> g_t$$

With $\gamma \neq 1$ the condition is $\hat{c} > \gamma \tilde{\sigma}^2$ as before

This characterization is robust to the specifics of the redistribution process

Main result: beyond usual summary statistics the precise shape of the left-tail of the wealth distribution matters

A possibility result



- Stationary wealth distributions yield plausible Gini values...
- ...may or may not support asset price bubbles
- Left-skew is critical

Stochastic deaths

Can a "perpetual youth" setting fix the aggregate "TVC"?

- If a new born household inherits the wealth of a specific decesead households

  $\hookrightarrow$ isomorphic to the baseline setting with permanent risk with discount factor $\hat{\rho} = \rho + \zeta$

- If wealth is pooled and redistributed equally, isomorphic to the model with jump risk

Stochastic deaths does not fix the aggregate "TVC": it continues to hinge on the balance between the time discount rate and the degree of idiosyncratic risk

Comparison (1): Constantinides-Duffie (1996)

Constantinides-Duffie (1996): permanent idiosyncratic risk raises the volatility of the pricing kernel→ higher risk premia.

$$\frac{c^i_{t+1}}{c^i_t} = \frac{c_{t+1}}{c_t} \exp\left(-\frac{1}{2}\widetilde{\sigma}^2_{t+1} + \widetilde{\sigma}_{t+1}\widetilde{\varepsilon}^i_{t+1}\right)$$

Key result: No-trade equilibrium: risk is permanent →agents face the same distribution of risk at all times→ symmetric buying pressure summarized by a single SDF

$$m^i_{t,t+1} = e^{-\rho}\left(\frac{c^i_{t+1}}{c^i_t}\right)^{-\gamma} = e^{-\rho}\left(\frac{c_{t+1}}{c_t}\right)^{-\gamma}\left(\frac{\eta^i_{t+1}}{\eta^i_t}\right)^{-\gamma}.$$

The present value of a claim to aggregate consumption is

$$\mathbb{E}_0\left[m^i_{0,t}c_t\right] = \mathbb{E}_0\left[\exp(-\rho t)\left(\frac{c_t}{c_0}\right)^{-1}\exp\left(\frac{t}{2}\widetilde{\sigma}^2 - \widetilde{\sigma}\sum_{k=1}^{t}\widetilde{\varepsilon}^i_k\right)c_t\right] = c_0\exp\left((\widetilde{\sigma}^2 - \rho)t\right)$$

If $\widetilde{\sigma}^2 > \rho$ the aggregate "TVC" fails and the no-trade equilibrium breaks down

Comparison (2): Huggett (1993)

The idea that $\uparrow \tilde{\sigma} \rightarrow \downarrow r_t^f$ is a key insight of the HA literature, Why does it not lead to asset price bubbles?

In typical HA settings, the income process is bounded away from zero:

$$\forall i, \forall \text{ states } y_i^t \geq \underline{y} > 0 \Longrightarrow \left(\eta_t^i\right)^{-\gamma} \leq \left(\underline{\eta}^{-\gamma}\right)$$

And therefore $\mathbb{E}_0\left[\int_0^1 \left(\eta_T^i\right)^{-\gamma}\right]$ stays bounded

Note however that with borrowing constraints $b_t^i \geq q_t^i$ the pricing equation is

$$m_t^i \geq \mathbb{E}_t\left[m_{t+1}^i(1+r)\right]$$

Therefore the IBC of an asset issuer is an inequality

$$b_t \geq \lim_{T\to\infty} \mathbb{E}_t\left[\sum_{k=1}^{T} \frac{m_{t+k}^i}{m_t^i} s_{t+k}\right] + \lim_{T\to\infty} \mathbb{E}_t\left[\frac{m_T^i}{m_t^i} b_T\right]$$

$\hookrightarrow$ Even if the "TVC" holds we can have $P > \mathbb{E}\left[PV(CF)\right]$

Comparison (3): BMS (2024)

Buy and Hold:

$$\frac{\mathcal{B}_0}{\mathcal{P}_0} = \lim_{T\to\infty}\left(\mathbb{E}\left[\int_0^T \bar{m}_t s_t K_t dt\right] + \mathbb{E}\left(\bar{m}_t \frac{\mathcal{B}_T}{\mathcal{P}_T}\right)\right)$$

Dynamic trading:

$$\begin{aligned}\frac{B_0}{P_0} &= \int\left(\mathbb{E}\left[\int_0^\infty m_t^i \cdot \eta_t^i s_t K_t\, dt\right] + \mathbb{E}\left[\int_0^\infty m_t^i \cdot \eta_t^i (1-\vartheta_t)^2 \chi^2 \frac{B_t}{P_t}\, dt\right]\right) di \\ &= \mathbb{E}\left[\int_0^\infty m_t^* s_t K_t\, dt\right] + \mathbb{E}\left[\int_0^\infty m_t^* (1-\vartheta_t)^2 \tilde{\sigma}_t^2 \frac{B_t}{P_t}\, dt\right]\end{aligned}$$

Service flow arising from retrading after idiosyncratic shock hits the capital returns:

$$\mu_t^{\Delta,i} = -\frac{s_t}{q_t^B}, \quad \tilde{\sigma}_t^{\Delta,i} = (1-\vartheta_t)\bar{\chi}\tilde{\sigma}_t$$

This paper: the key is that $\frac{d\eta_t^i}{\eta_t^i} = \tilde{\sigma}_t d\tilde{Z}_t^i \Longrightarrow \eta_t^i \to 0$ *a.s.*

Comments

- Offers a new perspective on the link between wealth inequality and asset pricing
- Re-interprets the "TVC" in a distributional perspective: the left tail of the wealth distribution matters
- Cool theoretical result: the aggregate "TVC" can fail even if the individual "TVC" holds

My contentions:

- If this is eventually a quantitative question, the model needs to have clear empirical counterparts
- Are poor/average households really marginal in pricing assets? Households rarely rebalance their portfolios and are very insensitive to price changes even at the top of the wealth distribution (Gabaix et al. 2024)
- Which wealth distribution does one need to look at then?
- Maybe the mechanism is more relevant if applied to broker institutions, but do they have the same SDF as their clients? Do they face the same constraints?

The qualitative model works perfectly to uncover the key mechanism, still it's unclear how much quantitative bite it can have

La Fin